

An improved feature selection method for classification on incomplete data: Non-negative latent factor-incorporated duplicate MIC

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ABSTRACT

Feature selection is an important data pre-processing procedure in the classification task, whose main purpose is to remove negligible/redundant features to reduce the computational cost, while improving the performance of the subsequent machine learning method. However, most feature selection methods can only work well with the complete data, but show poorly in the presence of the missing information, especially with a high missing rate, due probably to the severe irrelevance and redundancy of features. To address this issue, we proposed a novel feature selection method, namely non-negative latent factor incorporated duplicate maximal information coefficient (NLF-DMIC), which improves the effectiveness of feature selection for the classification of incomplete data. The NLF-DMIC method is fulfilled by the following three steps: (1) select category-friendly features by MIC based on “partial sample strategy” roughly; (2) use the NLF model to make an imputation for the missing data in terms of the filtered features; and (3) select features again by an improved maximal information coefficient (i.e., low-redundancy MIC, LMIC) method on the complete dataset. Finally, experiments on an artificially synthetic dataset and 8 real datasets show that the proposed NLF-DMIC method outperforms some state-of-the-art feature selection methods.

1. Introduction

Classification is one of the most common machine learning tasks, which allocates a class label to the samples by a specific algorithm. However, high dimensional data become common in the age of big data and increase the difficulty of classification. High dimensional data include image data, text data, genetic data and medical data, etc. These data contain a large number of unimportant features, which in turn increase the computational burden and even become a hindrance to machine learning (Ding et al., 2020). For example, there are always white pixels on the edge of handwritten digital pictures and these pixels are completely unimportant for the classification task.

Feature selection (FS) is an effective data preprocessing method to deal with the problem, which reduces dimensions by eliminating irrelevant features and redundant features. Due to the advantages of preventing overfitting, facilitating visualization, reducing storage space and computational cost, feature selection has a wide range of applications in text processing, genetic analysis, drug diagnosis, and other fields (Ghareb, Bakar, & Hamdan, 2016; Jain, Jain, & Jain, 2018; Lyu, Wan, Han, Liu, & Wang, 2017; Wei, Zhao, Feng, He, & Yu, 2020). In comparison to feature extraction, another method of dimensionality

reduction, feature selection remains the original features and therefore has better interpretability and readability (Li et al., 2017).

Depending on the combination with a certain model, feature selection methods can be divided into filter, wrapper and embedded. Filter, such as Pearson's correlation coefficient, Maximum Relevance Minimum Redundancy (mRMR) (Peng, Long, & Ding, 2005) and Relief (Reyes, Morell, & Ventura, 2015; Song, Si, Dai, & Yang, 2020), often selects features by revealing the correlation between variables. Wrapper, such as Support Vector Machine-Recursive Feature Elimination (SVM-RFE) (Guyon, Weston, Barnhill, & Vapnik, 2002), Genetic Algorithm (Amini & Hu, 2021; Chiang & Pell, 2004), selects features based on the classification accuracy of the feature subset. Embedded usually utilizes the idea of regularization to automatically select features during the training of the model (Hui & Hastie, 2005).

Wrapper and embedded tend to get higher accuracy, but have less computational efficiency and generalization ability. Filter tends to be simpler and has less computation because of its model-independence, thus is better for analysis of large-scale data. To obtain higher classification accuracy and computational efficiency, hybrid methods that combine different feature selection algorithms have been proposed.

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For example, Ghareb et al. combined a filter feature selection method with an enhanced Genetic Algorithm to improve categorization performance (Ghareb et al., 2016). Wei et al. combined the Gini index and maximal information coefficient to remove redundant features (Wei et al., 2020). Hybrid methods incorporate the advantages of different algorithms, which can greatly improve the efficiency of feature selection while achieving a high classification accuracy.

It is worth noting that most of the existing feature selection methods in terms of filtering are based on the assumption that the target dataset is complete. However, such an assumption is most likely to be invalid in practice, because the capacity of the information-acquiring apparatus is usually limited and the circumstance of the system is seriously complicated. Therefore, to realize dimensionality reduction on incomplete data, a general approach is a “two-step” strategy, which imputes the missing values firstly and selects features on the complete dataset.

A common imputation method in statistics is regression imputation, which predicts the missing value by establishing the regression model (Sefidian & Daneshpour, 2019; Stein & Kowalczyk, 2016). The main idea of KNN imputation is using k nearest neighbors to approximate incomplete samples (Huang & Lee, 2004; Pan, Yang, Cao, Lu, & Zhang, 2015). Matrix Factorization is one of the most effective methods of matrix completion, which recovers the missing data via decomposition and reconstruction of the matrix. For example, latent factor model is a low-rank matrix factorization model, which is more simple and efficient due to avoiding the pre-filling required by traditional imputation methods (Lee & Seung, 2000; Luo, Zhou, Li, et al., 2016; Luo, Zhou, Shang, Li, & Xia, 2016; Zhang, Wang, Ford, & Makedon, 2006).

Although this “two-step” strategy can select features on incomplete data, it raises another problem, that is, imputation increases the dependency of irrelevant features, which affects the subsequent feature selection and classification. Furthermore, since our purpose is to find out the best feature subset rather than to fill in all the missing values, “imputation then selection” will waste time on useless features. Another “two-step” method called PDS, is selecting features firstly, and then filling the missing values on the selected feature subset (Doquire & Verleyen, 2012). However, a problem with this approach is ignoring the fact that redundant features contribute to imputation but not to feature selection. In addition, it is difficult to ensure the effectiveness of feature selection with a high missing rate.

Considering the limitation of “two-step”, a new feature selection method based on “three-step” strategy is proposed, namely non-negative latent factor-incorporated duplicate maximal information coefficient (NLF-DMIC). Firstly, “partial sample strategy”-based MIC (PMIC) is used for “rough” feature selection to select category-friendly features. In this step, most irrelevant features are removed but redundant features are retained. Secondly, fill the missing values by the non-negative latent factor (NLF) model on the feature subset. The first feature selection greatly reduces the scale of the data matrix, so it is feasible to accelerate the learning of the model by matrix operation. Finally, select features again by low-redundancy MIC (LMIC) on the imputed data and obtain the optimal feature subset. LMIC can remove the irrelevant features and redundant features left by the first feature selection, so as to find a better feature subset. Our main contributions are:

1. The first “rough” feature selection based on PMIC eliminates most of the irrelevant features, which reduces the error and calculation time of subsequent imputation. PMIC incorporates maximal information coefficient with the “partial sample strategy” to select features from incomplete data.
2. A novel filter-based method LMIC is proposed to find a low-redundancy feature subset. LMIC uses maximal information coefficient (MIC) to calculate the correlation between variables instead of mutual information (MI), and finds the feature with maximum redundancy to avoid the impact of average.

3. We combine PMIC and LMIC to reduce the computational cost while achieving a high classification accuracy. Since irrelevant features are not conducive to imputation but redundant features contribute to imputation, we divide feature selection into two steps, which not only reduces the imputation errors, but also improves the efficiency of feature selection.
4. Three kinds of experiments comprehensively demonstrate the superiority of NLF-DMIC. To show the advantages of NLF-DMIC more intuitively, PMIC+NLF, LMIC and NLF-DMIC are respectively compared with other methods in the imputation task and classification task.

The remainder of this paper is organized as follows. Some well-known feature selection and imputation methods are introduced in Section 2. The details of the proposed NLF-DMIC are presented in Section 3. Then, experimental results and analysis are given in Section 4. Finally, the conclusions are drawn in Section 5.

2. Related works

In this section, we illustrate some well-known filter-based feature selection and imputation methods.

2.1. Feature selection

2.1.1. Mutual information

Entropy measures the uncertainty and average information of a random variable. The entropy of a random variable $X = (x_1, x_2, \dots, x_l)$ is defined as:

$$H(X) = - \sum_{i=1}^l p(x_i) \log p(x_i) \quad (1)$$

where $p(x_i)$ is the marginal probability of x_i .

Conditional entropy $H(X|Y)$ represents the uncertainty of X under the condition of variable $Y = (y_1, y_2, \dots, y_s)$, which is defined as:

$$H(X|Y) = - \sum_{i=1}^l \sum_{j=1}^s p(x_i, y_j) \log p(x_i|y_j) \quad (2)$$

where $p(x_i, y_j)$ is the joint probability of x_i and y_j , $p(x_i|y_j)$ is the conditional probability of x_i with respect to y_j .

Mutual information(MI) calculates the reduction of uncertainty of X under the condition of Y so as to measure the correlation between the two random variables. Compared with the correlation coefficient, MI is not limited to linear relationships and more general. The MI of random variable X and Y is defined as follows:

$$\begin{aligned} I(X; Y) &= H(X) - H(X|Y) \\ &= \sum_{x \in X} \sum_{y \in Y} p(x, y) \log \frac{p(x, y)}{p(x) \cdot p(y)} \end{aligned} \quad (3)$$

Given a variable Z , the conditional mutual information of X and Y under Z is defined as:

$$\begin{aligned} I(X; Y|Z) &= H(X|Z) - H(X|Y, Z) \\ &= \sum_{z \in Z} p(z) \sum_{x \in X} \sum_{y \in Y} p(x, y|z) \log \frac{p(x, y|z)}{p(x|z) \cdot p(y|z)} \end{aligned} \quad (4)$$

2.1.2. Maximal information coefficient

There are two problems with using mutual information directly for feature selection: firstly, MI is not normalized, so it is unreasonable to do a comparison on different datasets. Moreover, it is difficult to calculate on continuous variables. Maximal information coefficient (MIC) chooses the best discrete mode and is normalized to $[0, 1]$. The calculation of MIC is as follows (Reshef et al., 2011):

1. Given number a and b , grid the scatter graph formed by variables X and Y as a -by- b , and find out the largest $I_{a,b}^*(X; Y)$ in all grids.

2. Normalize $I_{a,b}^*(X; Y)$ to $[0, 1]$ by dividing $\log(\min\{a, b\})$
3. Select the largest $I_{a,b}^*(X; Y)$ as $MIC(X; Y)$ from all (a, b)

Thus,

$$MIC(X; Y) = \max_{a,b} \frac{I_{a,b}^*(X; Y)}{\log(\min\{a, b\})} \quad (5)$$

Reshef et al. compared the performance of MIC and Pearson, Spearman, MI based on Parzen window (Kwak & Choi, 2002), MI based on Kraskov estimation (Kraskov, Stögbauer, & Grassberger, 2004) and other correlation coefficients on 27 kinds of functional relationships (Reshef et al., 2011). According to the comparison results, methods such as mutual information estimators and maximal correlation show a strong preference for some functions, even at identical noise levels, which means they are unsuitable for identifying all potential relationships. MIC obtains values closed to 1 for most relationships, so it is more applicable.

2.1.3. Feature selection methods

MI-based feature selection (MIFS) generally follows the feature importance criteria:

$$J(f_i) = I(f_i; c) - \alpha \sum_{f_j \in S} I(f_i; f_j) + \beta \sum_{f_j \in S} I(f_i; f_j | c) \quad (6)$$

where $I(f_i; c)$ is the MI of feature variable f_i and class variable c ; $I(f_i; f_j)$ is the MI of feature variable f_i and f_j ; $I(f_i; f_j | c)$ is the conditional MI of f_i and f_j under c ; S is selected feature set; α and β are fixed values. The drawback of MIFS is that the second penalty term does not precisely characterize the feature redundancy because α and β are fixed values that cannot be adjusted with S .

Maximum Relevance Minimum Redundancy (mRMR) (Peng et al., 2005) is based on mutual information. mRMR selects features by the first-order incremental search method, which chooses the features with maximum relevancy and minimum redundancy at each iteration. The feature importance of mRMR is defined as:

$$J(f_i) = I(f_i; c) - \frac{1}{|S|} \sum_{f_j \in S} I(f_i; f_j) \quad (7)$$

where $|S|$ is the length of the set S . In mRMR, the weight of feature redundancy is adjusted with S , so that important features can be selected dynamically. Nevertheless, the disadvantage of mRMR is that averaging reduces the impact of those features that are severely redundant.

ReliefF (Reyes et al., 2015) is a distance-based feature selection method that iteratively updates the importance of each feature by using the distance between samples of the same class and samples of different classes on each feature. The feature weights are updated with the formula

$$W(f_i) = W(f_i) - \sum_{j=1}^k \frac{d(f_i, O, H_j)}{t \cdot k} + \sum_{c \neq \text{class}(O)} \frac{p(c)}{1 - p(\text{class}(O))} \sum_{j=1}^k \frac{d(f_i, O, M_j(c))}{t \cdot k} \quad (8)$$

where O is a randomly selected sample, $d(f_i, O, H_j)$, $d(f_i, O, M_j(c))$ are the distances on feature f_i between sample O and same-class sample H_j , different-class sample $M_j(c)$ respectively. t is the number of iterations and k is the number of nearest neighbors. $p(c)$ is the probability of class c , $\text{class}(O)$ is the class that O belongs to. The shortcoming of ReliefF is unable to wipe off redundant features.

2.2. Imputation

K-nearest neighbor (KNN) imputation is a nonparametric imputation technique, which defines a suitable distance and imputes missing values by means of KNN samples (Huang & Lee, 2004; Pan et al., 2015). For incomplete data, there are two strategies for calculating the

sample distance. One is simply filling the data, like mean substitution, and then calculating the distance on the complete data. The other is “partial distance” strategy, which calculates the distance by the available values, and ignores the missing values. The distance between sample x_i and x_j is

$$d(x_i, x_j) = \sqrt{\frac{m}{\sum_{r=1}^m I_r} \sum_{r=1}^m (x_i^{(r)} - x_j^{(r)})^2 \cdot I_r} \quad (9)$$

where

$$I_r = \begin{cases} 1, & \text{if } x_i^{(r)} \neq \text{null and } x_j^{(r)} \neq \text{null} \\ 0, & \text{otherwise} \end{cases} \quad (10)$$

KNN imputation has the advantage of being easy implemented and avoiding computational time on learning model, but is not easy to calculate the similarity matrix of high dimensional data. Moreover, at a high missing rate, it is difficult to estimate the distance in either pre-filling or “partial distance”.

Regression imputation is a common imputation method (Sefidian & Daneshpour, 2019; Stein & Kowalczyk, 2016). During the regression, the missing attribute is regarded as dependent variable, other attributes as independent variables. The predicted values by the regression model are regarded as the filling values. Common regression models include Linear Regression, Lasso Regression, Support Vector Regression, etc. The main advantage of these methods is to exploit the relationships of features to predict missing values. However, the performance is unsatisfactory when there is no relationship between independent variables and dependent variables.

Matrix Factorization is an effective means of matrix completion. A decomposition model is defined and learned by the observed records, and the complete data are obtained by matrix reconstruction. Matrix Factorization models include Singular Value Decomposition (SVD), Latent Factor Model (LFM) (Lee & Seung, 2000), symmetric and non-negative latent factor (SNLF) (Song, Li, Luo, Yang, & Wang, 2019), etc. LFM is widely used due to its simpleness, small computation and no pre-filling. The model of LFM is

$$R_{ij} = (PQ)_{ij} \quad (11)$$

where R_{ij} is an element of matrix $R_{n \times m}$. P and Q are the latent factor matrices.

To learn P and Q , a loss function is built as:

$$L(P, Q) = \sum_{R_{ij} \neq \text{null}} (R_{ij} - (PQ)_{ij})^2 \quad (12)$$

The optimal P and Q can be found by minimizing the above loss function, and then the imputed values can be obtained by matrix reconstruction. Gradient descent is a simple technique to optimize parameters, but convergence is slow. Conjugate gradient has faster convergence, but is more complicated to implement than gradient descent. Lee et al. proposed “multiplicative update rules” to balance speed and ease of implementation by updating P and Q alternately (Lee & Seung, 2000). Luo et al. proposed a regularized single-element-based NLF model to achieve high computational efficiency and prediction accuracy (Luo, Zhou, Xia, & Zhu, 2014).

3. Proposed method

3.1. PMIC

Generally, feature selection does not work on missing data, so imputation is needed beforehand. However, considering irrelevant features severely affect imputation, we use MIC to select features on missing data by “partial sample strategy” (PSS), which is called PMIC. “Partial sample strategy” means using the available values of all feature variables and class variable to calculate MIC, that is

$$MIC_{PSS}(X; Y) = MIC(X_{obs}; Y_{obs}) \quad (13)$$

where X_{obs} and Y_{obs} are the values in X and Y which are observed simultaneously.

A necessary condition for Eq. (13) is that the data are missing at random (MAR). MAR means that the variable X or its distribution does not depend on its missing values X_{miss} . Another assumption is that samples are sufficient. In fact, we can make these assumptions since MAR is common in reality and high dimensional data are more accessible. The Law of Large Numbers states that when the samples are sufficient, the sample distribution is almost close to the real distribution. In this case, $P(X) \approx P(X_{obs})$, $P(X, Y) \approx P(X_{obs}, Y_{obs})$. Therefore, we can use $MIC_{PSS}(X; Y)$ to approximate $MIC(X; Y)$.

By using $MIC_{PSS}(X; Y)$ to exploit the correlation between feature and class, the feature importance of PMIC can be defined as:

$$PMIC(f_i) = MIC_{PSS}(f_i; c) \quad (14)$$

3.2. MIC-based low redundancy feature selection

Although mRMR considers feature redundancy, its calculation by taking the average is not satisfactory. In fact, we pay more attention to the maximum of pairwise feature redundancy rather than the average. So, we propose a new MIC-based low-redundancy feature selection method (LMIC), whose feature importance is defined as follows:

$$LMIC(f_i) = MIC(f_i; c) \times (1 - \max_{f_j \in S} \{MIC(f_i; f_j)\}) \quad (15)$$

$MIC(f_i; c)$ is the correlation between feature and class label, representing the feature importance of f_i ; $MIC(f_i; f_j)$ is the correlation between feature f_i and f_j , representing the feature redundancy. Obviously, larger $MIC(f_i; c)$ and smaller $\max_{f_j \in S} \{MIC(f_i; f_j)\}$, more important f_i is.

In addition, because MIC ranges from 0 to 1, we utilize multiplication to increase the influence of redundancy, instead of traditional summation. Apparently, if one of the selected features has a strong correlation with the candidate feature f_i ($MIC(f_i; f_j)$ close to 1), the $LMIC(f_i)$ will be close to 0. Therefore, the impact of feature redundancy is magnified.

3.3. Non-negative latent factor

Considering the loss function of LFM:

$$\begin{aligned} L(P, Q) &= \frac{1}{2} \sum_{R_{ij} \neq null} (R_{ij} - (PQ)_{ij})^2 \\ &= \frac{1}{2} (\|I \odot (R - PQ)\|_F^2) \end{aligned} \quad (16)$$

where $\odot$ is Hadamard product and I is a indicator matrix:

$$I_{ij} = \begin{cases} 1, & \text{if } R_{ij} \neq null \\ 0, & \text{otherwise} \end{cases} \quad (17)$$

To avoid overfitting, we add regularization:

$$\begin{aligned} L(P, Q) &= \frac{1}{2} (\|I \odot (R - PQ)\|_F^2 \\ &\quad + \lambda_1 \|P\|_F^2 + \lambda_2 \|Q\|_F^2) \end{aligned} \quad (18)$$

where λ_1 and λ_2 are regularization factors.

To minimize the above loss function, the partial derivative of L with respect to P and Q is

$$\begin{aligned} \frac{\partial L}{\partial P} &= -(I \odot (R - PQ) \odot I)Q^T + \lambda_1 P \\ &= -(I \odot R)Q^T + (I \odot PQ)Q^T + \lambda_1 P \\ \frac{\partial L}{\partial Q} &= -P^T(I \odot (R - PQ) \odot I) + \lambda_2 Q \\ &= -P^T(I \odot R) + P^T(I \odot PQ) + \lambda_2 Q \end{aligned} \quad (19)$$

With gradient decent, each parameter is obtained via the following updates:

$$\begin{aligned} P &= P + \alpha_1 \odot ((I \odot R)Q^T - (I \odot PQ)Q^T - \lambda_1 P) \\ Q &= Q + \alpha_2 \odot (P^T(I \odot R) - P^T(I \odot PQ) - \lambda_2 Q) \end{aligned} \quad (20)$$

where α_1 and α_2 are the learning factor matrices with the same size as P and Q .

In fact, most data matrices are non-negative, thus non-negative updates of P and Q not only ensure the nonnegativity of the target matrix, but also reserve the physical significance of matrix factorization. For this purpose, let

$$\begin{aligned} \alpha_1 &= \frac{P}{(I \odot PQ)Q^T + \lambda_1 P} \\ \alpha_2 &= \frac{Q}{P^T(I \odot PQ) + \lambda_2 Q} \end{aligned} \quad (21)$$

it should be noted that the a_{ik} of α_1 is equal to $P_{ik}/((I \odot PQ)Q^T + \lambda_1 P)_{ik}$, so is α_2 . Then we have

$$\begin{aligned} P &= P + \alpha_1 \odot ((I \odot R)Q^T - (I \odot PQ)Q^T - \lambda_1 P) \\ &= P \odot \frac{(I \odot R)Q^T}{(I \odot PQ)Q^T + \lambda_1 P} \end{aligned} \quad (22)$$

$$\begin{aligned} Q &= Q + \alpha_2 \odot (P^T(I \odot R) - P^T(I \odot PQ) - \lambda_2 Q) \\ &= Q \odot \frac{P^T(I \odot R)}{P^T(I \odot PQ) + \lambda_2 Q} \end{aligned} \quad (23)$$

So, when the initial $P^{(0)}$ and $Q^{(0)}$ are non-negative, each iterated $P^{(i)}$ and $Q^{(i)}$ are also non-negative. Here, in order to speed up the convergence and fully utilize the computer for parallel computation, P and Q are updated alternately in matrix form.

3.4. NLF-DMIC

To reduce the imputation errors and computational cost, we propose a new “three-step” feature selection method, named non-negative latent factor-incorporated duplicate maximal information coefficient (NLF-DMIC). Firstly, calculate the PMIC between each feature and the class, and obtain a feature subset according to the feature score. Secondly, complete the data matrix by NLF model on the selected feature subset. Finally, select features again by LMIC on the complete dataset, and get the optimal feature subset. Fig. 1 depicts the NLF-DMIC feature selection based on “three-step” strategy.

Step 1: Utilize PMIC to roughly select features on incomplete data. Given the target feature number m , regard each feature as variable x , class label as variable c , calculate $PMIC(f_i)$ of each feature according to (14) and retain the top m features. Details are shown in Algorithm 1. This filter method eliminates most of the irrelevant features, but reserves some redundant features. In fact, irrelevant features increase imputation errors while redundant features make for imputation. Moreover, without calculating feature redundancy, the computational cost is lower.

Algorithm 1 PMIC-based Feature Selection

Input: Incomplete dataset R_{origin} with feature set $F = \{f_1, f_2, \dots, f_M\}$ and class label c , target feature number m

- 1: Initialize $S_1 = \emptyset$
- 2: **for** $i = 1$ to M **do**
- 3: Calculate each feature importance $PMIC(f_i)$ on R_{origin} according to (14)
- 4: Sort the features in descending order according to $PMIC(f_i)$
- 5: Select top m features and add them to S_1

Output: Selected feature subset S_1

Step 2: Utilize NLF model to impute missing values on feature subset. Initiate P , Q , λ_1 , λ_2 and max iteration t . Alternately update P and Q according to (22) and (23) until the convergence conditions are reached, refactor data matrix by learned P, Q . Details are shown in Algorithm 2. Because NLF model is established on the feature subset with few irrelevant features, the imputation errors and computational time are both reduced.

Step 3: Utilize LMIC to select features again on the imputed data, and obtain the optimal feature subset. Given a candidate feature set

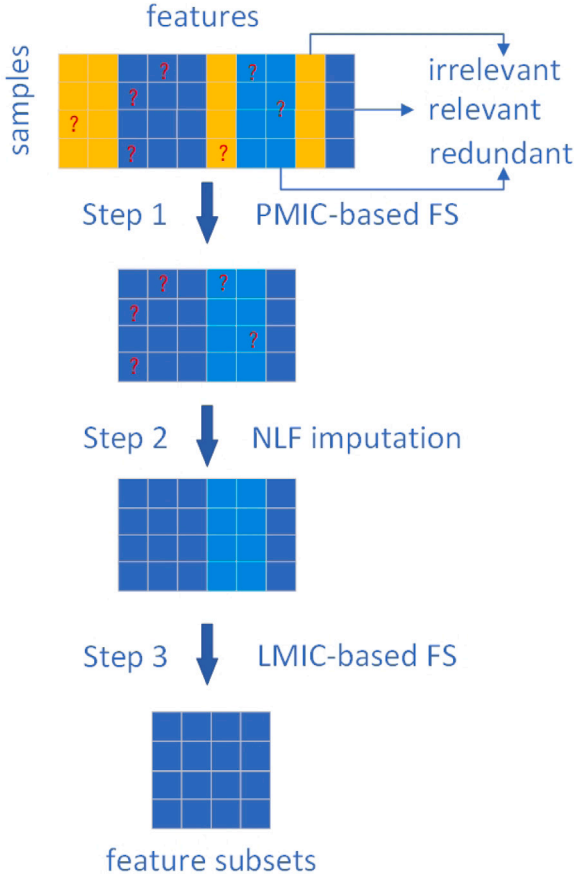


Fig. 1. NLF-DMIC feature selection based on "three-step" strategy.

Algorithm 2 Non-negative Latent Factor

Input: Incomplete data matrix $R_{n \times m}$, rank d , λ_1 , λ_2 , max iteration t

- 1: Initialize non-negative matrix $P_{n \times d}$ randomly
- 2: Initialize non-negative matrix $Q_{d \times m}$ randomly
- 3: Initialize I according to (17)
- 4: Initialize $iter = 0$
- 5: **while** not converge and $iter < t$ **do**
- 6: Update P according to (22)
- 7: Update Q according to (23)
- 8: $iter = iter + 1$
- 9: Impute R by (11) and obtain R_{cpl}

Output: Complete data R_{cpl}

$S_1 = \{f_1, f_2, \dots, f_m\}$ and the target feature number h ($h < m$). Initialize selected feature set $S = \emptyset$, calculate the feature importance of each feature according to (15), and find the best feature f_{best} . Add f_{best} to S and delete f_{best} from S_1 . Repeat it until all features are sorted. Details are shown in Algorithm 3. As mentioned in 3.2, LMIC is superior in removing redundant features. Therefore, a large number of redundant features reserved in Step 1 are eliminated in this step by LMIC.

In summary, considering that irrelevant features are not conducive to imputation while redundant features contribute to imputation, NLF-DMIC divides feature selection into two steps, respectively before and after imputation. The first feature selection only removes the irrelevant features by PMIC, so NLF can make full use of the effective information in redundant features for imputation. Meanwhile, the efficiency of imputation has been significantly improved after the first feature selection. The second feature selection removes the remaining irrelevant features and redundant features, and then finds a better feature

Algorithm 3 LMIC-based Feature Selection

Input: Complete dataset R_{cpl} with feature set $S_1 = \{f_1, f_2, \dots, f_m\}$ and class label c , target feature number h

- 1: Initialize $S = \emptyset$, $iter = 0$
- 2: **for** $i = 1$ to m **do**
- 3: Calculate $MIC(f_i, c)$ according to (5)
- 4: $LMIC(f_i) = MIC(f_i, c)$
- 5: Select the feature f_{best} with the largest $LMIC(f_i)$
- 6: $S = S \cup \{f_{best}\}$; $S_1 = S_1 - \{f_{best}\}$
- 7: $iter = iter + 1$
- 8: **while** $iter < h$ **do**
- 9: **for** f_i in S_1 **do**
- 10: **for** f_j in S **do**
- 11: Calculate $MIC(f_i, f_j)$ according to (5)
- 12: Update $LMIC(f_i)$ according to (15)
- 13: Select the feature f_{best} with the largest $LMIC(f_i)$
- 14: $S = S \cup \{f_{best}\}$; $S_1 = S_1 - \{f_{best}\}$
- 15: $iter = iter + 1$

Output: Selected feature subset S

subset. In addition, in comparison to the methods only considering the correlation between feature and class, those calculating feature redundancy will increase the computational cost, such as mRMR and LMIC. Therefore, the combination of PMIC and LMIC further reduces the computation.

3.5. Time complexity

Given n samples, each sample consists of M features $\{f_1, f_2, \dots, f_M\}$ and 1 label c . Given v_{ab} grids of size (a, b) , the time complexity of MIC is approximately $O(v_{ab} \cdot (a \cdot b + n))$ according to (5). Given the first target feature number m , the time complexity of PMIC-based FS is approximately $O(v_{ab} \cdot (a \cdot b + n) \cdot M)$ according to Algorithm 1. Given the rank d and iteration t , the time complexity of NLF is approximately $O(t \cdot n \cdot m \cdot d)$ according to Algorithm 2. Given the target feature number h , the time complexity of LMIC-based FS is approximately $O(h^2 \cdot (3m - 2h) \cdot v_{ab} \cdot (a \cdot b + n))$ according to Algorithm 3. In summary, since $a \cdot b \ll n$, the time complexity of NLF-DMIC is approximately $O((h^2 \cdot (3m - 2h) + M) \cdot v_{ab} \cdot n + t \cdot n \cdot m \cdot d)$.

In addition, the time complexity of MIFS, mRMR, and ReliefF can be obtained according to (3), (7), and (8). Assuming that each feature has v_f different values and the class c has v_c different values, the time complexity of MIFS is approximately $O(M \cdot (v_f \cdot v_c + n))$, and that of mRMR is $O(h^2 \cdot (3M - 2h) \cdot (v_f^2 + n))$. Given the iteration t , the time complexity of ReliefF is approximately $O(t \cdot M^2 \cdot n)$.

4. Experiments

The experiments are divided into three main parts. Firstly, to verify the first feature selection can efficiently reduce imputation errors, we compared PMIC+NLF with other 3 imputation methods on a synthetic dataset and 8 real datasets. KNN imputation is a simple, effective and parameter-free method. Regression imputation is often the method of choice in data preprocessing. NLF is a matrix complementation method based on matrix decomposition techniques and is widely used for prediction of sparse scoring matrices in recommender systems. Secondly, to demonstrate "three-step" method NLF-DMIC can obtain better feature subset, we compared NLF-DMIC ("three-step"), NLF+LMIC ("two-step") and PDS ("two-step") (Doquire & Verleysen, 2012) on 8 real datasets. At last, to demonstrate the advantage of low redundancy, the proposed LMIC was compared with MIFS, MICFS, mRMR and ReliefF on 8 real datasets. Since LMIC is a filter feature selection method, these state-of-the-art filter algorithms are chosen for comparison. All experiments were implemented in Python with the help of sklearn and statsmodels (Pedregosa et al., 2011).

Table 1
Datasets description.

Datasets	Samples	Features	Classes	Category balanced
Synthetic	5000	300	2	yes
Madelon	4400	500	2	yes
PD	756	754	2	yes
LSVT	126	309	2	yes
QSAR	1687	1024	2	yes
CNAE-9	1080	857	9	yes
Mfeat	2000	649	10	yes
Yale	165	1024	15	yes
Lung	203	3312	5	yes

Table 2
Hyperparameters of NLF for each dataset.

Datasets	m	d	λ_1	λ_2
Synthetic	50	10	0.01	0.01
Madelon	50	6	0.01	0.02
PD	150	10	0.01	0.001
LSVT	150	10	0.05	0.01
QSAR	300	20	0.01	0.01
CNAE-9	300	20	0.01	0.01
Mfeat	400	20	0.005	0.001
Yale	500	20	0.005	0.01
Lung	1000	20	0.01	0.01

Table 3
RMSE for LR, KNN, NLF and PMIC+NLF on a Synthetic Dataset and 8 Real Datasets.

Datasets	Missing rate(%)	LR	KNN	NLF	PMIC+NLF
Synthetic	30	0.0631 $\pm$ 0.0002	0.0719 $\pm$ 0.0003	0.0641 $\pm$ 0.0002	0.0607 $\pm$ 0.0001
	40	0.0649 $\pm$ 0.0003	0.0734 $\pm$ 0.0002	0.0642 $\pm$ 0.0001	0.0609 $\pm$ 0.0001
	50	0.0983 $\pm$ 0.0048	0.0753 $\pm$ 0.0003	0.0645 $\pm$ 0.0001	0.0613 $\pm$ 0.0001
	60	0.1177 $\pm$ 0.0038	0.0784 $\pm$ 0.0003	0.0648 $\pm$ 0.0001	0.0621 $\pm$ 0.0001
Madelon	30	0.1527 $\pm$ 0.0013	0.1529 $\pm$ 0.0010	0.1471 $\pm$ 0.0010	0.1318 $\pm$ 0.0014
	40	0.1483 $\pm$ 0.0014	0.1537 $\pm$ 0.0006	0.1466 $\pm$ 0.0013	0.1338 $\pm$ 0.0021
	50	0.1466 $\pm$ 0.0011	0.1547 $\pm$ 0.0008	0.1464 $\pm$ 0.0010	0.1374 $\pm$ 0.0026
	60	0.1452 $\pm$ 0.0010	0.1550 $\pm$ 0.0010	0.1462 $\pm$ 0.0013	0.1425 $\pm$ 0.0018
PD	30	0.1588 $\pm$ 0.0120	0.0724 $\pm$ 0.0010	0.0689 $\pm$ 0.0008	0.0673 $\pm$ 0.0010
	40	0.1421 $\pm$ 0.0071	0.0758 $\pm$ 0.0009	0.0698 $\pm$ 0.0013	0.0686 $\pm$ 0.0014
	50	0.1385 $\pm$ 0.0059	0.0793 $\pm$ 0.0010	0.0717 $\pm$ 0.0009	0.0717 $\pm$ 0.0011
	60	0.1355 $\pm$ 0.0076	0.0845 $\pm$ 0.0015	0.0752 $\pm$ 0.0009	0.0757 $\pm$ 0.0009
LSVT	30	0.2314 $\pm$ 0.0411	0.1133 $\pm$ 0.0031	0.1118 $\pm$ 0.0039	0.1111 $\pm$ 0.0041
	40	0.2260 $\pm$ 0.0318	0.1176 $\pm$ 0.0037	0.1131 $\pm$ 0.0030	0.1127 $\pm$ 0.0030
	50	0.2240 $\pm$ 0.0167	0.1230 $\pm$ 0.0033	0.1151 $\pm$ 0.0024	0.1158 $\pm$ 0.0027
	60	0.2182 $\pm$ 0.0161	0.1287 $\pm$ 0.0029	0.1193 $\pm$ 0.0033	0.1209 $\pm$ 0.0037
QSAR	30	0.2162 $\pm$ 0.0034	0.2335 $\pm$ 0.0015	0.2367 $\pm$ 0.0016	0.2305 $\pm$ 0.0018
	40	0.3569 $\pm$ 0.0703	0.2409 $\pm$ 0.0021	0.2367 $\pm$ 0.0019	0.2318 $\pm$ 0.0021
	50	0.2597 $\pm$ 0.0069	0.2501 $\pm$ 0.0022	0.2376 $\pm$ 0.0023	0.2355 $\pm$ 0.0023
	60	0.2503 $\pm$ 0.0138	0.2613 $\pm$ 0.0025	0.2380 $\pm$ 0.0025	0.2398 $\pm$ 0.0028
CNAE-9	30	0.0907 $\pm$ 0.0027	0.1266 $\pm$ 0.0022	0.0848 $\pm$ 0.0010	0.0819 $\pm$ 0.0010
	40	0.0867 $\pm$ 0.0025	0.1303 $\pm$ 0.0014	0.0840 $\pm$ 0.0009	0.0814 $\pm$ 0.0009
	50	0.0860 $\pm$ 0.0019	0.1340 $\pm$ 0.0015	0.0833 $\pm$ 0.0007	0.0811 $\pm$ 0.0007
	60	0.0850 $\pm$ 0.0019	0.1373 $\pm$ 0.0011	0.0827 $\pm$ 0.0008	0.0811 $\pm$ 0.0008
Mfeat	30	0.2150 $\pm$ 0.0172	0.1927 $\pm$ 0.0008	0.1604 $\pm$ 0.0009	0.1540 $\pm$ 0.0009
	40	0.3553 $\pm$ 0.0038	0.2008 $\pm$ 0.0008	0.1616 $\pm$ 0.0014	0.1559 $\pm$ 0.0015
	50	0.3782 $\pm$ 0.0014	0.2118 $\pm$ 0.0009	0.1632 $\pm$ 0.0013	0.1593 $\pm$ 0.0017
	60	0.3845 $\pm$ 0.0008	0.2241 $\pm$ 0.0010	0.1653 $\pm$ 0.0016	0.1644 $\pm$ 0.0020
Yale	30	0.1253 $\pm$ 0.0013	0.1633 $\pm$ 0.0015	0.1258 $\pm$ 0.0009	0.1240 $\pm$ 0.0012
	40	0.1348 $\pm$ 0.0011	0.1680 $\pm$ 0.0012	0.1279 $\pm$ 0.0015	0.1272 $\pm$ 0.0016
	50	0.1398 $\pm$ 0.0008	0.1704 $\pm$ 0.0008	0.1326 $\pm$ 0.0010	0.1330 $\pm$ 0.0011
	60	0.1455 $\pm$ 0.0005	0.1735 $\pm$ 0.0008	0.1394 $\pm$ 0.0008	0.1398 $\pm$ 0.0007
Lung	30	0.1191 $\pm$ 0.0011	0.1413 $\pm$ 0.0006	0.1271 $\pm$ 0.0008	0.1275 $\pm$ 0.0007
	40	0.1245 $\pm$ 0.0008	0.1444 $\pm$ 0.0009	0.1285 $\pm$ 0.0009	0.1292 $\pm$ 0.0009
	50	0.1297 $\pm$ 0.0009	0.1480 $\pm$ 0.0009	0.1304 $\pm$ 0.0008	0.1317 $\pm$ 0.0007
	60	0.1357 $\pm$ 0.0009	0.1522 $\pm$ 0.0011	0.1335 $\pm$ 0.0011	0.1355 $\pm$ 0.0011

4.1. Datasets

4.1.1. Synthetic dataset

To prove that irrelevant features increase imputation errors, a synthetic dataset with 5000 samples is generated for the first experiment. Each sample has 300 features and a class label. The top 50 are relevant features and the last 250 are irrelevant features. The class label is 0 or 1. In detail, all 300 variables subject to Gaussian distribution with mean in $\{1, 2, 3\}$ and variance in $\{1, 2, 3\}$. The covariance between any two relevant features is 0.8, and the others are 0. The labels for each sample are generated as follows:

$$temp_i = \sum_{j=1}^{50} x_i^{(j)} \quad i = 1, 2, \dots, 5000$$

$$c_i = \begin{cases} 1, & \text{if } temp_i - \overline{temp} \geq 0 \\ 0, & \text{otherwise} \end{cases} \quad (24)$$

where $\overline{temp}$ is the mean of $temp_i$, $i = 1, 2, \dots, 5000$.

4.1.2. Real datasets

8 real datasets were employed in our experiments: Madelon, PD, LSVT, QSAR, CNAE-9, Multiple Features (Mfeat), Yale and Lung. These datasets can be found in [UCIDatasets](#) and [ASUDatasets](#). These datasets are all category-balanced and other descriptions are shown in [Table 1](#).

Table 4
MAE for LR, KNN, NLF and PMIC+NLF on a Synthetic Dataset and 8 Real Datasets.

Datasets	Missing rate(%)	LR	KNN	NLF	PMIC+NLF
Synthetic	30	0.0503 ± 0.0002	0.0573 ± 0.0002	0.0511 ± 0.0001	0.0483 ± 0.0001
	40	0.0517 ± 0.0002	0.0584 ± 0.0001	0.0512 ± 0.0001	0.0486 ± 0.0001
	50	0.0740 ± 0.0038	0.0599 ± 0.0002	0.0513 ± 0.0001	0.0489 ± 0.0001
	60	0.0908 ± 0.0034	0.0623 ± 0.0002	0.0516 ± 0.0001	0.0495 ± 0.0001
Madelon	30	0.1200 ± 0.0012	0.1213 ± 0.0008	0.1184 ± 0.0010	0.1024 ± 0.0013
	40	0.1172 ± 0.0012	0.1220 ± 0.0005	0.1180 ± 0.0012	0.1044 ± 0.0019
	50	0.1161 ± 0.0008	0.1229 ± 0.0007	0.1177 ± 0.0010	0.1075 ± 0.0024
	60	0.1152 ± 0.0008	0.1231 ± 0.0008	0.1175 ± 0.0012	0.1120 ± 0.0017
PD	30	0.0727 ± 0.0012	0.0397 ± 0.0007	0.0400 ± 0.0006	0.0375 ± 0.0006
	40	0.0686 ± 0.0013	0.0418 ± 0.0006	0.0404 ± 0.0008	0.0382 ± 0.0008
	50	0.0670 ± 0.0018	0.0437 ± 0.0006	0.0416 ± 0.0006	0.0404 ± 0.0006
	60	0.0658 ± 0.0017	0.0470 ± 0.0011	0.0434 ± 0.0007	0.0431 ± 0.0007
LSVT	30	0.0671 ± 0.0057	0.0519 ± 0.0022	0.0642 ± 0.0020	0.0637 ± 0.0022
	40	0.0718 ± 0.0054	0.0553 ± 0.0027	0.0652 ± 0.0020	0.0650 ± 0.0018
	50	0.0754 ± 0.0044	0.0591 ± 0.0026	0.0663 ± 0.0017	0.0669 ± 0.0017
	60	0.0781 ± 0.0047	0.0626 ± 0.0021	0.0684 ± 0.0027	0.0696 ± 0.0030
QSAR	30	0.0930 ± 0.0027	0.0876 ± 0.0011	0.1297 ± 0.0018	0.1210 ± 0.0018
	40	0.1593 ± 0.0076	0.0927 ± 0.0018	0.1291 ± 0.0020	0.1210 ± 0.0022
	50	0.1458 ± 0.0033	0.0996 ± 0.0020	0.1294 ± 0.0023	0.1228 ± 0.0025
	60	0.1342 ± 0.0036	0.1076 ± 0.0024	0.1289 ± 0.0027	0.1241 ± 0.0029
CNAE-9	30	0.0236 ± 0.0006	0.0387 ± 0.0008	0.0120 ± 0.0002	0.0130 ± 0.0002
	40	0.0237 ± 0.0008	0.0405 ± 0.0007	0.0118 ± 0.0001	0.0129 ± 0.0002
	50	0.0246 ± 0.0008	0.0422 ± 0.0008	0.0117 ± 0.0001	0.0128 ± 0.0002
	60	0.0236 ± 0.0009	0.0438 ± 0.0007	0.0116 ± 0.0001	0.0127 ± 0.0002
Mfeat	30	0.1577 ± 0.0135	0.1069 ± 0.0004	0.1130 ± 0.0006	0.1064 ± 0.0008
	40	0.2981 ± 0.0045	0.1130 ± 0.0005	0.1140 ± 0.0012	0.1076 ± 0.0011
	50	0.3263 ± 0.0018	0.1220 ± 0.0006	0.1150 ± 0.0009	0.1094 ± 0.0012
	60	0.3332 ± 0.0012	0.1326 ± 0.0008	0.1166 ± 0.0012	0.1125 ± 0.0014
Yale	30	0.0895 ± 0.0009	0.1169 ± 0.0010	0.0923 ± 0.0007	0.0897 ± 0.0009
	40	0.0968 ± 0.0007	0.1215 ± 0.0010	0.0939 ± 0.0012	0.0922 ± 0.0014
	50	0.1013 ± 0.0006	0.1251 ± 0.0007	0.0977 ± 0.0007	0.0972 ± 0.0009
	60	0.1063 ± 0.0004	0.1292 ± 0.0006	0.1030 ± 0.0006	0.1030 ± 0.0006
Lung	30	0.0873 ± 0.0006	0.1044 ± 0.0004	0.0956 ± 0.0006	0.0957 ± 0.0005
	40	0.0914 ± 0.0005	0.1068 ± 0.0006	0.0966 ± 0.0006	0.0970 ± 0.0006
	50	0.0954 ± 0.0005	0.1097 ± 0.0007	0.0982 ± 0.0007	0.0990 ± 0.0006
	60	0.1000 ± 0.0006	0.1129 ± 0.0007	0.1006 ± 0.0008	0.1020 ± 0.0008

- Madelon: It is an artificial dataset. The 5 original dimensions constitute 5 informative features. 15 linear combinations of the 5 features are added together to constitute 20 redundant features. One must divide the instances into 2 classes based on these 20 features. The other 480 features are called “probes” and have no predictive capability (Guyon, Gunn, Benhur, & Dror, 2004).
- PD: The data were collected from 188 patients, 107 males and 81 females, aged between 33 and 87, from the Department of Neurology, CerrahpaÅYa Faculty of Medicine, Istanbul University. The control group consisted of 64 healthy people (23 males and 41 females) with ages ranging from 41 to 82 (Sakar et al., 2019).
- LSVT: The dataset was created by Athanasios Tsanas of the University of Oxford. There are 126 samples from 14 individuals with 309 features. Each feature corresponds to the implementation of a speech signal processing algorithm, which aims to characterize the signal objectively (Tsanas, Little, Fox, & Ramig, 2013).
- QSAR: The dataset was used for developing classification QSAR models. Properties (molecular fingerprints) were computed at the Milano Chemometrics and QSAR Research Group on a group of chemicals provided by the US National Center of Computational Toxicology (Grisoni, Consonni, & Ballabio, 2019).
- CNAE-9: This dataset contains 1080 free-text business description documents of Brazilian companies, grouped into a subset of 9 classes in a table called “National Classification of Economic Activities”.
- Mfeat: This is a dataset containing features of handwritten digits (0–9) extracted from a set of Dutch utility maps. 200 patterns per category have been converted into binary images (Dua & Graff, 2017).
- Yale: It contains 165 grayscale images of 15 people in GIF format. Each subject has 11 images and each image has a different facial expression: center-light, left-light, right-light, happy, sad, wearing glasses, no glasses, wink, surprised, sleepy and normal (Cai, He, Hu, Han, & Huang, 2007).
- Lung: This dataset consists of 203 lung images. Each image contains 3312 pixels, which are regarded as features. The samples are separated into 5 classes, which are 5 different types of lung.

4.1.3. Data pre-processing

To eliminate the effect of scale, values of each attribute were transformed into [0, 1] by Min–Max normalization. The conversion formula is

$$\hat{f}_j^{(i)} = \frac{f_j^{(i)} - \min(f_j)}{\max(f_j) - \min(f_j)} \quad (25)$$

where $f_j^{(i)}$ is the i_{th} dimension of feature vector f_j .

To compare the performance of different methods, in the first two groups of experiments, missing values were randomly inserted into those complete datasets. The pattern is missing completely at random,

Table 5
Classification accuracy(%) for NLF and PMIC+NLF on a Synthetic Dataset and 8 Real Datasets.

Datasets	Missing rate(%)	SVM		ADB		RF	
		NLF	PMIC+NLF	NLF	PMIC+NLF	NLF	PMIC+NLF
Synthetic	30	97.36 ± 0.38	98.23 ± 0.31	93.93 ± 0.72	94.73 ± 0.59	98.19 ± 0.31	98.22 ± 0.33
	40	97.07 ± 0.35	97.73 ± 0.36	94.54 ± 0.59	95.10 ± 0.64	97.78 ± 0.24	97.82 ± 0.32
	50	96.98 ± 0.37	97.45 ± 0.32	94.72 ± 0.61	95.28 ± 0.40	97.50 ± 0.38	97.53 ± 0.32
	60	96.43 ± 0.55	96.91 ± 0.43	94.45 ± 0.67	94.95 ± 0.62	96.80 ± 0.48	96.90 ± 0.43
Madelon	30	57.45 ± 1.54	61.18 ± 1.71	58.75 ± 1.94	65.33 ± 3.14	61.16 ± 2.16	68.11 ± 1.78
	40	56.46 ± 1.98	59.78 ± 1.60	57.35 ± 2.20	61.64 ± 2.12	60.01 ± 1.62	65.67 ± 1.93
	50	56.11 ± 2.30	59.04 ± 2.00	55.56 ± 1.73	58.14 ± 2.71	59.32 ± 2.06	63.75 ± 1.55
	60	55.89 ± 1.99	58.56 ± 1.95	54.44 ± 2.35	55.60 ± 1.99	58.12 ± 1.87	60.98 ± 1.78
PD	30	83.14 ± 2.73	85.01 ± 2.27	77.02 ± 3.06	79.05 ± 2.27	85.70 ± 2.33	86.49 ± 2.12
	40	82.80 ± 2.39	84.10 ± 2.22	75.52 ± 2.56	77.67 ± 2.95	84.95 ± 2.00	85.27 ± 2.19
	50	82.44 ± 2.14	84.24 ± 1.96	76.27 ± 2.70	76.51 ± 2.70	84.74 ± 2.71	85.15 ± 2.36
	60	81.95 ± 2.21	83.27 ± 2.25	75.21 ± 2.25	75.42 ± 2.90	84.76 ± 2.08	85.01 ± 1.85
LSVT	30	77.02 ± 9.24	82.54 ± 6.28	71.84 ± 5.99	74.47 ± 8.06	80.88 ± 5.17	82.54 ± 3.75
	40	73.95 ± 10.07	78.16 ± 7.91	72.63 ± 7.03	73.86 ± 7.47	80.61 ± 6.24	82.63 ± 5.38
	50	72.81 ± 8.97	77.37 ± 7.07	70.61 ± 9.48	74.04 ± 5.07	79.56 ± 6.43	81.14 ± 6.18
	60	69.47 ± 9.49	73.95 ± 10.19	69.04 ± 7.40	71.75 ± 7.50	79.47 ± 6.46	82.46 ± 6.23
QSAR	30	89.86 ± 0.97	90.03 ± 0.96	83.45 ± 1.62	84.39 ± 1.59	89.43 ± 0.96	89.96 ± 0.93
	40	90.35 ± 1.04	90.40 ± 1.00	83.65 ± 1.45	84.21 ± 1.23	90.01 ± 1.01	90.30 ± 0.98
	50	90.31 ± 1.03	90.47 ± 0.98	83.20 ± 1.64	84.15 ± 0.97	90.20 ± 1.01	90.34 ± 1.06
	60	90.43 ± 1.02	90.42 ± 1.09	83.21 ± 1.59	84.52 ± 1.28	90.30 ± 1.04	90.35 ± 1.04
CNAE-9	30	80.06 ± 2.26	82.02 ± 2.60	64.57 ± 2.97	69.88 ± 2.78	70.78 ± 2.93	77.01 ± 2.76
	40	73.97 ± 2.21	75.83 ± 1.97	59.67 ± 3.66	63.12 ± 3.05	65.12 ± 2.95	72.73 ± 2.43
	50	73.97 ± 2.21	75.83 ± 1.97	59.67 ± 3.66	63.12 ± 3.05	65.12 ± 2.95	72.73 ± 2.43
	60	60.83 ± 2.52	63.82 ± 2.46	51.50 ± 3.46	55.58 ± 2.07	57.20 ± 2.91	63.27 ± 1.64
Mfeat	30	98.10 ± 0.45	97.48 ± 0.59	89.27 ± 1.23	89.56 ± 1.54	97.91 ± 0.49	97.76 ± 0.63
	40	97.98 ± 0.57	97.16 ± 0.82	88.51 ± 1.39	88.43 ± 1.26	97.52 ± 0.60	97.15 ± 0.79
	50	97.64 ± 0.50	96.60 ± 0.66	87.39 ± 1.32	87.42 ± 1.54	96.93 ± 0.56	96.49 ± 0.72
	60	97.66 ± 0.71	96.22 ± 0.68	86.74 ± 1.67	86.17 ± 1.65	96.82 ± 0.68	96.04 ± 0.76
Yale	30	58.40 ± 5.37	62.13 ± 6.62	38.60 ± 5.90	42.73 ± 7.00	68.80 ± 6.32	69.60 ± 5.74
	40	58.27 ± 6.14	60.13 ± 4.93	36.80 ± 7.23	38.67 ± 6.24	68.20 ± 5.83	69.47 ± 5.41
	50	57.67 ± 5.78	60.67 ± 5.93	38.00 ± 6.01	41.13 ± 9.48	66.93 ± 6.96	68.00 ± 4.98
	60	55.20 ± 5.79	56.87 ± 4.97	34.93 ± 6.05	37.53 ± 6.86	65.27 ± 5.98	65.87 ± 5.87
Lung	30	93.99 ± 2.87	95.90 ± 2.68	81.97 ± 5.00	82.57 ± 4.55	92.57 ± 3.44	94.26 ± 3.10
	40	92.51 ± 3.67	95.52 ± 2.62	82.30 ± 4.74	84.21 ± 4.55	90.87 ± 4.21	93.88 ± 3.73
	50	93.17 ± 2.73	95.25 ± 2.60	81.91 ± 3.93	83.66 ± 4.76	91.97 ± 3.61	93.77 ± 3.29
	60	92.95 ± 3.37	94.64 ± 2.95	83.93 ± 4.88	84.86 ± 5.51	91.80 ± 3.36	93.66 ± 3.75

i.e. MCAR, and each feature may contain missing values. To test the effect of missing rate, 4 levels of missing rate were set, respectively 30%, 40%, 50% and 60%. To mitigate randomness, each experiment with different missing rates was repeated 30 times and experimental results were shown as mean±standard deviation.

4.2. Evaluation and setting of hyper-parameters

Evaluation: In the first experiment, we mainly care about the closeness of imputations to the actual values, thus Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE) were used to measure imputation errors. In the last two experiments, we chose classification accuracy (ACC) to evaluate the feature subsets. ACC, RMSE and MAE are shown as follows:

$$\begin{aligned}
 ACC &= \frac{T}{N} \\
 RMSE &= \sqrt{\frac{\sum_{R_{ij} \in V} (R_{ij} - \hat{R}_{ij})^2}{|V|}} \\
 MAE &= \frac{\sum_{R_{ij} \in V} |R_{ij} - \hat{R}_{ij}|_{abs}}{|V|}
 \end{aligned} \quad (26)$$

where T is the number of samples which are classified correctly from the test set and N is the number of total samples in the test set for the

classifier evaluation. $\hat{R}_{ij}$ is the predicted value, R_{ij} is the true value, V is the missing value set, $|V|$ is the length of V , $|\cdot|_{abs}$ is absolute value.

Setting of hyper-parameters (Table 2):

- For m in Algorithm 1, since redundant features help imputation and only irrelevant features are expected to be removed, m can be set slightly larger in this step. After a large number of testing experiments, m is set as 50, 50, 150, 150, 300, 300, 400, 500 and 1000 respectively for Synthetic, Madelon, PD, LSVT, QSAR, CNAE-9, Mfeat, Yale and Lung.
- d, λ_1, λ_2 in Algorithm 2 are trained by 5-fold cross-validation. The parameters with the lowest mean value of RMSE are selected. Firstly, we fix λ_1 and λ_2 , select the best d by one-dimensional search; then, fix d , select λ_1 and λ_2 by grid search. For example, in Madelon, $d = 6$, $\lambda_1 = 0.01$, $\lambda_2 = 0.02$. The training process is shown in Figs. 2 and 3.
- For h in Algorithm 3, it should not be too large, since we prefer to choose a relatively small feature subset, and we do not assign a fixed value to h in case this value favors a particular method. The value of h varies from 1 to 50(100) and the optimal feature subsets will be selected from the top h features with the highest accuracy.

Table 6
Classification accuracy(%) for NLF-DMIC, NLF+LMIC and PDS on 8 real datasets by SVM.

Datasets	Missing rate(%)	NLF-DMIC		NLF+LMIC		PDS	
		Best ACC	No. F	Best ACC	No. F	Best ACC	No. F
Madelon	30	68.73 $\pm$ 1.69	15	67.11 $\pm$ 1.57	25	73.60 $\pm$ 3.14	15
	40	67.48 $\pm$ 1.87	10	65.60 $\pm$ 0.90	10	67.97 $\pm$ 2.33	20
	50	66.58 $\pm$ 2.27	15	63.85 $\pm$ 1.83	40	63.98 $\pm$ 1.90	20
	60	60.86 $\pm$ 1.96	45	59.03 $\pm$ 1.73	20	60.58 $\pm$ 1.82	15
PD	30	85.36 $\pm$ 2.22	40	83.49 $\pm$ 2.71	5	80.88 $\pm$ 3.37	40
	40	84.96 $\pm$ 2.16	30	83.57 $\pm$ 2.13	45	79.94 $\pm$ 2.99	50
	50	84.80 $\pm$ 2.12	50	84.05 $\pm$ 2.11	25	78.88 $\pm$ 2.80	45
	60	82.48 $\pm$ 2.37	45	83.73 $\pm$ 1.93	25	78.66 $\pm$ 3.19	50
LSVT	30	90.70 $\pm$ 3.95	10	89.65 $\pm$ 4.64	15	74.56 $\pm$ 9.68	35
	40	87.46 $\pm$ 5.38	50	85.53 $\pm$ 4.03	10	72.81 $\pm$ 8.22	50
	50	86.40 $\pm$ 5.04	25	88.86 $\pm$ 3.56	5	72.81 $\pm$ 8.19	15
	60	88.16 $\pm$ 4.84	5	82.89 $\pm$ 5.28	25	71.75 $\pm$ 7.60	30
QSAR	30	90.38 $\pm$ 1.09	35	90.47 $\pm$ 1.09	50	90.40 $\pm$ 1.12	35
	40	89.96 $\pm$ 0.77	50	89.77 $\pm$ 1.11	50	90.25 $\pm$ 1.16	35
	50	90.31 $\pm$ 1.14	50	90.05 $\pm$ 1.14	50	90.07 $\pm$ 0.90	45
	60	89.89 $\pm$ 0.88	5	90.02 $\pm$ 0.86	5	89.91 $\pm$ 1.33	40
Mfeat	30	98.10 $\pm$ 0.46	100	97.93 $\pm$ 0.52	65	97.01 $\pm$ 0.84	100
	40	98.11 $\pm$ 0.53	90	98.02 $\pm$ 0.52	100	95.63 $\pm$ 0.95	100
	50	97.38 $\pm$ 0.46	85	97.90 $\pm$ 0.48	90	93.47 $\pm$ 1.15	100
	60	97.00 $\pm$ 0.53	80	97.08 $\pm$ 0.61	60	88.39 $\pm$ 1.64	100
CNAR-9	30	76.99 $\pm$ 1.57	100	74.79 $\pm$ 1.36	100	79.88 $\pm$ 1.61	100
	40	72.23 $\pm$ 2.27	100	66.10 $\pm$ 2.25	100	74.87 $\pm$ 2.93	100
	50	63.86 $\pm$ 1.97	100	62.96 $\pm$ 1.96	100	68.93 $\pm$ 2.49	100
	60	54.66 $\pm$ 2.26	100	55.49 $\pm$ 2.38	100	61.37 $\pm$ 2.88	90
Yale	30	63.93 $\pm$ 7.16	30	59.47 $\pm$ 6.65	30	42.93 $\pm$ 5.58	50
	40	60.47 $\pm$ 7.39	50	63.07 $\pm$ 6.97	35	41.07 $\pm$ 6.49	45
	50	64.93 $\pm$ 6.59	25	68.47 $\pm$ 5.79	30	38.60 $\pm$ 6.79	45
	60	61.33 $\pm$ 6.46	45	61.53 $\pm$ 6.10	50	33.80 $\pm$ 7.34	50
Lung	30	96.17 $\pm$ 3.09	50	95.96 $\pm$ 3.28	20	89.13 $\pm$ 4.34	50
	40	95.74 $\pm$ 2.79	35	95.63 $\pm$ 3.05	40	87.27 $\pm$ 4.21	50
	50	96.34 $\pm$ 3.19	25	94.97 $\pm$ 2.93	35	87.21 $\pm$ 3.79	50
	60	96.67 $\pm$ 1.96	15	96.12 $\pm$ 2.38	25	85.14 $\pm$ 4.15	50

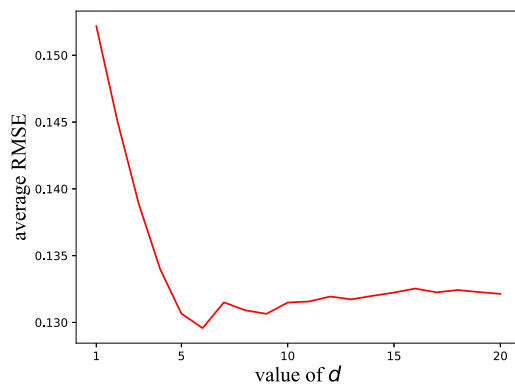


Fig. 2. NLF model with different values of d on Madelon.

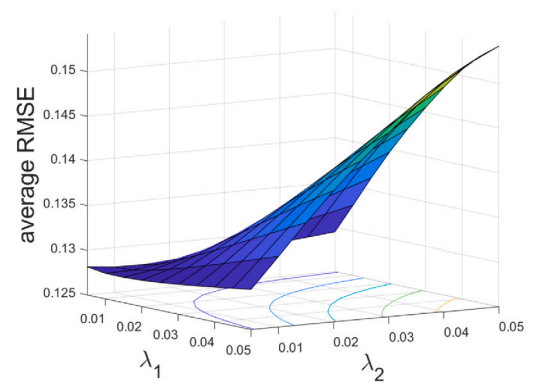


Fig. 3. NLF model with different values of λ_1 and λ_2 on Madelon.

4.3. Experimental results and analysis

At first, we compared PMIC+NLF, NLF, LR and KNN ($k=5$) imputation on a synthetic dataset and 8 real datasets. PMIC+NLF refers to PMIC-based FS before imputation by NLF. Since the number of imputed values by two different strategies is not the same (NLF provides the predictions for all missing values, while PMIC+NLF only gives predictions for relevant features), we only took the same fillings for comparison. In detail, the m features selected by PMIC are used to calculate the imputation error RMSE and MAE. For PMIC+NLF and NLF, we took the RMSE and MAE after convergence for comparison. The convergences of

PMIC+NLF and NLF on Madelon are shown as an example in Fig. 4. To prove the m selected features are “good”, three classifiers were used to test the imputed datasets, namely Support Vector Machine(SVM), Adaboost(ADB) and Random Forest(RF). The experimental results are provided in Tables 3–5.

Tables 3 and 4 indicate that PMIC+NLF has almost lowest RMSE on synthetic dataset, Madelon, PD, LSVT, CNAE-9, Mfeat and Yale, and lowest MAE on synthetic dataset, Madelon, PD, Mfeat and Yale. In particular, PMIC+NLF performs better on most datasets when compared to NLF only, which means the first FS indeed reduces imputation errors. Meanwhile, as Table 5 shows, the classification accuracy of PMIC+NLF

Table 7
Classification accuracy(%) for NLF-DMIC, NLF+LMIC and PDS on 8 real datasets by ADB.

Datasets	Missing rate(%)	NLF-DMIC		NLF+LMIC		PDS	
		Best ACC	No. F	Best ACC	No. F	Best ACC	No. F
Madelon	30	62.61 $\pm$ 1.23	40	59.58 $\pm$ 2.62	20	68.34 $\pm$ 2.84	15
	40	63.20 $\pm$ 0.170	15	59.11 $\pm$ 2.15	20	62.63 $\pm$ 2.36	15
	50	59.08 $\pm$ 0.95	10	56.35 $\pm$ 2.28	25	57.47 $\pm$ 2.15	35
	60	55.87 $\pm$ 1.66	50	53.90 $\pm$ 2.10	5	0.55.73 $\pm$ 1.73	10
PD	30	79.72 $\pm$ 2.82	15	80.09 $\pm$ 2.40	5	76.28 $\pm$ 3.12	45
	40	78.90 $\pm$ 1.84	10	78.33 $\pm$ 2.57	20	75.11 $\pm$ 2.86	50
	50	77.75 $\pm$ 2.75	20	76.56 $\pm$ 2.88	50	73.25 $\pm$ 3.19	50
	60	77.84 $\pm$ 2.32	35	76.95 $\pm$ 2.67	15	71.40 $\pm$ 2.82	45
LSVT	30	79.82 $\pm$ 5.19	5	85.61 $\pm$ 5.19	5	72.37 $\pm$ 7.24	30
	40	76.93 $\pm$ 4.42	40	75.79 $\pm$ 6.67	5	70.70 $\pm$ 7.37	50
	50	82.54 $\pm$ 3.83	5	76.58 $\pm$ 4.27	5	70.18 $\pm$ 6.96	45
	60	79.56 $\pm$ 4.79	20	74.82 $\pm$ 6.56	45	70.00 $\pm$ 7.00	30
QSAR	30	87.02 $\pm$ 1.29	5	87.07 $\pm$ 1.29	5	87.18 $\pm$ 1.65	5
	40	86.26 $\pm$ 1.17	5	86.03 $\pm$ 1.29	5	87.23 $\pm$ 1.89	5
	50	86.72 $\pm$ 1.21	5	85.10 $\pm$ 1.21	5	87.44 $\pm$ 1.45	5
	60	86.25 $\pm$ 0.79	5	85.17 $\pm$ 0.79	5	87.42 $\pm$ 1.37	5
Mfeat	30	89.86 $\pm$ 0.53	30	89.65 $\pm$ 0.43	45	83.94 $\pm$ 1.93	100
	40	89.14 $\pm$ 0.58	40	89.86 $\pm$ 0.58	35	82.19 $\pm$ 2.51	100
	50	88.03 $\pm$.53	45	89.01 $\pm$ 0.52	40	76.49 $\pm$ 2.93	100
	60	86.51 $\pm$ 0.71	50	87.27 $\pm$ 0.71	45	64.76 $\pm$ 3.17	100
CNAR-9	30	69.27 $\pm$ 1.39	100	68.33 $\pm$ 1.36	100	67.90 $\pm$ 2.37	90
	40	66.15 $\pm$ 2.27	100	64.17 $\pm$ 2.32	100	61.85 $\pm$ 2.98	75
	50	61.53 $\pm$ 1.96	100	60.85 $\pm$ 2.07	100	55.26 $\pm$ 2.89	90
	60	55.00 $\pm$ 2.39	100	54.08 $\pm$ 2.35	75	48.41 $\pm$ 2.56	55
Yale	30	42.53 $\pm$ 7.16	30	46.87 $\pm$ 5.63	50	36.13 $\pm$ 6.97	50
	40	42.60 $\pm$ 7.51	35	45.13 $\pm$ 6.79	25	33.47 $\pm$ 7.41	50
	50	47.13 $\pm$ 6.93	20	44.00 $\pm$ 7.36	15	28.80 $\pm$ 5.14	45
	60	48.40 $\pm$ 6.04	25	40.87 $\pm$ 6.26	15	23.07 $\pm$ 5.63	50
Lung	30	85.30 $\pm$ 3.32	30	86.01 $\pm$ 3.47	40	78.47 $\pm$ 5.44	40
	40	87.76 $\pm$ 3.45	5	86.17 $\pm$ 3.45	10	79.40 $\pm$ 5.28	50
	50	86.89 $\pm$ 3.29	50	86.89 $\pm$ 2.54	35	76.72 $\pm$ 4.48	50
	60	88.42 $\pm$ 2.89	5	88.25 $\pm$ 2.25	25	73.77 $\pm$ 6.31	50

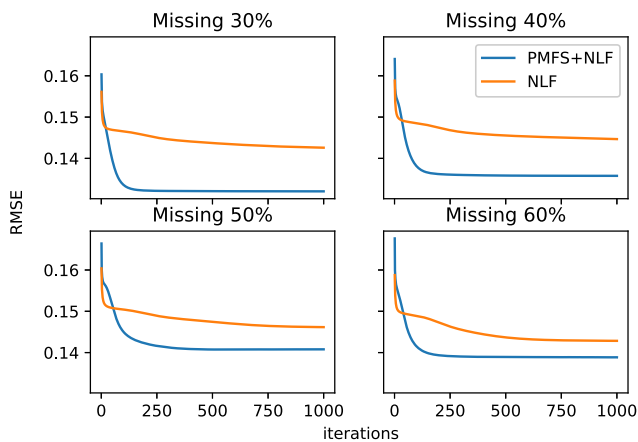


Fig. 4. Convergence of PMIC+NLF and NLF at different missing rates on Madelon.

is better or no worse than NLF on most of the datasets, which validates the efficiency of FS. In addition, although KNN and LR imputation sometimes perform better than PMIC+NLF (RMSE on lung, MAE on LSVT and QSAR), their computational cost is much higher than NLF with the increasing samples and features. However, PMIC+NLF is not robust at high missing rates. For example, PMIC+NLF performs poorly on PD, LSVT and Yale when the missing rate is higher than 50%. A possible reason is that high missing rate will interfere the capability of “partial sample strategy” to estimate data distribution.

Secondly, to validate the superiority of the “three-step” strategy, we compared our “three-step” method NLF-DMIC with two “two-step” methods NLF+LMIC and PDS (Doquire & Verleysen, 2012) on the 8 real datasets. NLF-DMIC includes PMIC-based FS, NLF imputation and LMIC-based FS. NLF+LMIC consists of NLF imputation and LMIC-based FS. PDS selects features by MIC based on “partial distance” firstly, then fills the missing values by KNN imputation. To assess the validity of the combined feature selection method, we respectively used SVM, ADB and RF to test the feature subsets obtained by different methods. Each subset was split into 70% training set and 30% test set. Tables 6–8 show the best classification accuracy of top h features on test set by different classifiers. No. F is the number of features corresponding to the best accuracy.

As Tables 6–8 show, in 32 classification tests of SVM, ADB and RF, NLF-DMIC obtains 17,18,21 times highest accuracy respectively. The comparison results of NLF-DMIC and NLF+LMIC on 8 real datasets fully demonstrated that the first “rough” FS, as the only difference between these two methods, actually reduced the imputation errors to some extent. PDS performs poorly in most cases, especially at high missing rates. This is because PDS selects features only once, so a large portion of the features need to be removed before imputation, resulting in missing information. In contrast, the second FS of NLF-DMIC can correct the first rough FS and thus achieve better classification results. Moreover, the average number of features for NLF-DMIC, NLF+LMIC and PDS on all datasets is 42, 40 and 51, respectively. NLF-DMIC is not the method with the least number of features, which is because the experiment prefers classification accuracy. In fact, a few more features are acceptable since the optimal accuracy is only searched within 50 (100) features.

Table 8
Classification accuracy(%) for NLF-DMIC, NLF+LMIC and PDS on 8 real datasets by RF.

Datasets	Missing rate(%)	NLF-DMIC		NLF+LMIC		PDS	
		Best ACC	No. F	Best ACC	No. F	Best ACC	No. F
Madelon	30	67.81 ± 1.95	20	64.21 ± 1.72	20	76.64 ± 3.39	15
	40	65.93 ± 1.10	10	61.63 ± 0.58	25	70.64 ± 2.45	20
	50	65.76 ± 1.92	45	61.78 ± 1.22	15	65.00 ± 2.34	25
	60	62.16 ± 1.82	30	60.83 ± 1.93	40	60.08 ± 1.87	30
PD	30	86.51 ± 2.56	25	85.84 ± 2.79	20	83.42 ± 2.49	40
	40	86.30 ± 2.38	30	85.59 ± 2.22	20	82.45 ± 2.59	50
	50	85.46 ± 1.98	30	84.61 ± 2.01	35	81.75 ± 2.82	50
	60	84.33 ± 2.15	45	84.60 ± 1.90	35	80.78 ± 2.46	50
LSVT	30	88.51 ± 5.30	10	88.68 ± 4.60	5	81.14 ± 5.91	40
	40	87.98 ± 4.81	15	87.02 ± 4.22	35	80.18 ± 6.29	40
	50	89.04 ± 3.83	5	87.98 ± 4.37	20	79.30 ± 5.89	50
	60	87.46 ± 6.01	35	85.26 ± 6.01	35	80.79 ± 8.09	45
QSAR	30	90.66 ± 1.05	25	90.83 ± 1.08	25	90.22 ± 1.18	50
	40	90.17 ± 1.11	50	90.08 ± 1.09	40	90.18 ± 1.19	35
	50	90.58 ± 1.07	45	90.58 ± 1.07	45	90.07 ± 0.87	45
	60	90.18 ± 0.89	45	90.16 ± 0.91	35	89.92 ± 1.36	40
Mfeat	30	98.25 ± 0.54	100	98.12 ± 0.69	100	95.97 ± 0.95	100
	40	97.97 ± 0.49	95	97.74 ± 0.52	95	94.47 ± 1.22	100
	50	97.33 ± 0.46	85	97.87 ± 0.53	95	91.48 ± 1.26	100
	60	96.22 ± 0.76	45	96.60 ± 0.71	50	85.06 ± 1.84	100
CNAR-9	30	77.72 ± 1.50	100	75.31 ± 1.39	100	76.50 ± 2.53	100
	40	72.96 ± 2.24	100	71.63 ± 1.91	100	71.38 ± 2.58	75
	50	70.51 ± 1.97	100	69.19 ± 1.88	100	64.95 ± 2.88	90
	60	62.71 ± 2.39	100	61.03 ± 2.30	100	58.45 ± 2.71	90
Yale	30	69.53 ± 5.63	50	71.33 ± 5.20	50	53.00 ± 6.53	50
	40	67.20 ± 7.12	50	68.47 ± 7.65	35	49.73 ± 8.28	50
	50	72.07 ± 6.35	35	74.00 ± 6.84	50	45.67 ± 7.03	50
	60	68.33 ± 6.52	45	67.27 ± 6.69	35	38.13 ± 6.08	50
Lung	30	94.21 ± 3.35	50	94.70 ± 3.16	45	87.05 ± 4.11	50
	40	95.19 ± 2.96	35	94.64 ± 3.11	30	86.12 ± 4.15	50
	50	95.03 ± 3.16	30	93.88 ± 3.16	30	85.63 ± 4.08	50
	60	95.46 ± 1.97	50	94.15 ± 2.43	10	83.22 ± 3.57	50

Thirdly, in order to verify the low redundancy of LMIC, LMIC was compared with MIFS (Lewis, 1992), MICFS (Reshef et al., 2011), mRMR (Peng et al., 2005) and ReliefF (Reyes et al., 2015) on 8 real datasets. MICFS is a replacement of MI with MIC in MIFS. At first, feature subsets were obtained from the top h features ranked by different FS methods. Then, classifiers were used to evaluate the quality of feature subsets. In this group of experiments, the 8 datasets are all complete and not inserted with missing values. To better characterize the effect of the number of features, the value of h varies from 1 to 50 (100). In addition, to demonstrate the generalization of the filter methods, three different classifiers are used to evaluate the feature subsets, respectively SVM, ADB and RF. The experimental results in Fig. 5 are the classification accuracy of the test set.

We can conclude from Fig. 5 that on most datasets, LMIC achieves high classification accuracy at the early stage of feature selection, which means LMIC is able to eliminate redundant features effectively and find the better feature combination. On PD, CNAE, Mfeat, Yale and Lung, LMIC shows a greater advantage in accuracy and feature number. Taking the dataset Yale and classifier RF as an example, LMIC achieves the classification accuracy of 0.8 with only the first 9 features, which is much higher than the maximum value of 0.72 in other methods. On LSVT and QSAR, ReliefF, mRMR and MICFS have their own advantages on different classifiers, but in general, they have similar results. On Madelon, LMIC has the highest classification accuracy on 5 features but is unsatisfactory when the feature number is more than 5. In fact, there are only 5 relevant features and 15 redundant features in Madelon (written in dataset descriptions). Therefore, after the selection of the

five features, due to the characteristic of low redundancy, LMIC selects other irrelevant features, which reduces the classification accuracy.

Finally, in order to investigate whether there is a significant difference between PMIC+NLF and other methods, statistical significance test, Wilcoxon sum-rank test, is conducted on the imputation results, consisting of RMSE and MAE values from 30 randomized experiments for each algorithm. Wilcoxon sum-rank test is non-parametric and widely used to compare two algorithms (Carrasco, García, Rueda, Das, & Herrera, 2020). The null hypothesis of the test is that the two algorithms perform similarly. The test results are listed in Table 9. The symbol “*” and “**” respectively represent that differences are significant at 95% (p -value < 0.05) and 99% (p -value < 0.01), while “-” indicates that there is no significant difference between two algorithms.

Table 9 shows that the PMIC+NLF has a very significant difference with the peer algorithms on most datasets with different missing rates in terms of the RMSE and MAE. It illustrates that the proposed PMIC+NLF is significantly better than other compared methods in most cases. However, on dataset LSVT with scale of 126×309 , the superiority of PMIC+NLF is not significantly better than the traditional NLF. This shows that the new PMIC+NLF is appropriate to process large scale incomplete data.

4.4. Computational time analysis

As mentioned in the contributions, NLF-DMIC has an advantage in computational time compared to the “two-step” strategy. To prove this, Madelon, Lung and QSAR have been used to test the computational

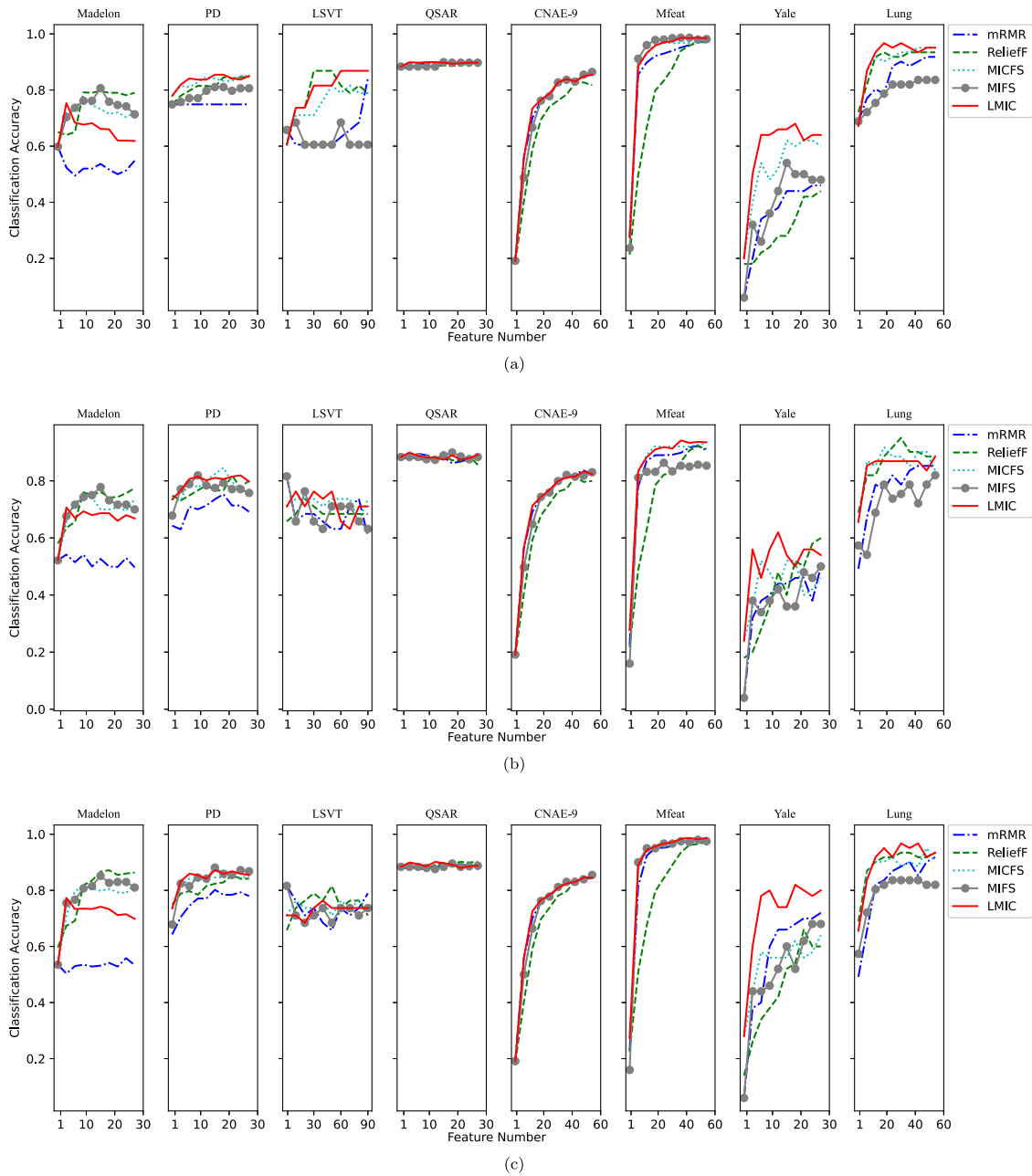


Fig. 5. Comparison of LMIC to other filter methods on 8 real datasets (a) Classification on SVM (b) Classification on ADB (c) Classification on RF.

time. In Madelon, the number of samples is much more than the number of features. In Lung, the number of samples is much less than the number of features. In QSAR, they are similar. The experimental results are in seconds.

Table 10 gives the time consumption of NLF-DMIC and NLF+LMIC. Because PMIC-based FS only considers the correlation with the class and does not calculate the redundancy between features, it takes almost less than a second by parallel computation. Through the first FS, the computational time of NLF and LMIC has been significantly reduced. In the case of Madelon, the entire NLF-DMIC process takes 221 s, which is only about 3/10 of the NLF+LMIC. Total time consumption shows the superiority of NLF-DMIC.

In addition, Table 11 shows the time consumption of LR, KNN, NLF, and PMIC+NLF. From the table, we can find that the efficiency of regression and KNN imputation is higher when data matrix is small, while PMIC+NLF is instead the most efficient when the data matrix becomes large. This is because the computation of NLF is related to the

number of iterations. The computational time of NLF is more affected by the iterations when the data matrix is small, while it is less affected by the iterations when the data matrix is large. It is worth noting that the computational times for PMIC+NLF and NLF in Table 11 are lower than those in Table 10. This is because in Table 10 the number of iterations is fixed at 500, while in Table 11 the iterations stop earlier due to convergence.

Table 12 shows the time consumption of MIFS, MICFS, mRMR, ReliefF, and LMIC. From the table, we can find that MIFS and MICFS are the most efficient because they do not calculate the redundancy between features. ReliefF is the least efficient because it requires constant iteration and calculation of distance. LMIC and mRMR are in between because they consider the redundancy of features. The computation of MI is faster on Madelon, but MIC is faster on Lung and QSAR, which may be related to the distribution of the data. In summary, the computational time of LMIC is acceptable, especially when a large

Table 9
Wilcoxon sum-rank test for RMSE and MAE results.

Datasets	Missing rate(%)	RMSE			MAE		
		PMIC+NLF vs. LR	PMIC+NLF vs. KNN	PMIC+NLF vs. NLF	PMIC+NLF vs. LR	PMIC+NLF vs. KNN	PMIC+NLF vs. NLF
Synthetic	30	**	**	**	**	**	**
	40	**	**	**	**	**	**
	50	**	**	**	**	**	**
	60	**	**	**	**	**	**
Madelon	30	**	**	**	**	**	**
	40	**	**	**	**	**	**
	50	**	**	**	**	**	**
	60	**	**	**	**	**	**
PD	30	**	**	**	**	**	**
	40	**	**	**	**	**	**
	50	**	**	—	**	**	**
	60	**	**	*	**	**	—
LSVT	30	**	*	—	**	**	—
	40	**	**	—	**	**	—
	50	**	**	—	**	**	—
	60	**	**	*	**	**	*
QSAR	30	**	**	**	**	**	**
	40	**	**	**	**	**	**
	50	**	**	**	**	**	**
	60	**	**	**	**	**	**
CNAE-9	30	**	**	**	**	**	**
	40	**	**	**	**	**	**
	50	**	**	**	**	*	**
	60	**	**	**	**	**	**
Mfeat	30	**	**	**	**	**	**
	40	**	**	**	**	**	**
	50	**	**	**	**	**	**
	60	**	**	*	**	**	**
Yale	30	**	**	**	—	**	**
	40	**	**	—	**	**	**
	50	**	**	—	**	**	**
	60	**	**	*	**	**	—
Lung	30	**	**	*	**	**	—
	40	**	**	**	**	**	**
	50	**	**	**	**	**	**
	60	—	**	**	**	**	**

Table 10
Time consumption (seconds) of NLF-DMIC and NLF+LMIC on 3 datasets.

Dataset	Samples	Features	NLF+LMIC			NLF-DMIC			
			NLF	LMIC	Total	PMIC	NLF	LMIC	Total
Madelon	4400	500	37	749	786	1	15	205	221
Lung	203	3312	29	231	260	1	11	69	81
Qsar	1687	1024	67	1010	1077	1	27	259	287

Table 11
Time consumption (seconds) of LR, KNN, NLF, and PMIC+NLF on 3 datasets.

Datasets	Samples	Features	LR	KNN	NLF	PMIC+NLF
Madelon	4400	500	9.4	7.2	20.2	11.9
Lung	203	3312	32.7	1.1	14.4	1.0
Qsar	1687	1024	54.5	11.3	24.3	6.1

Table 12
Time consumption(seconds) of MIFS, MICFS, mRMR, ReliefF, and LMIC on 3 datasets.

Datasets	Samples	Features	MIFS	MICFS	mRMR	ReliefF	LMIC
Madelon	4400	500	0.4	1.0	25	501.8	262.4
Lung	203	3312	2.2	0.9	111.7	628.5	121.2
Qsar	1687	1024	0.8	0.3	40	823.1	15.4

number of features have been removed by the first feature selection (PMIC-based FS).

5. Conclusion

The “curse of dimensionality” and data missing are the common problems in data mining at the current era. To achieve dimensionality reduction of high dimensional incomplete data, a traditional “two-step” method is filling the missing values firstly and then feature selection. However, the large number of irrelevant features increases imputation errors and time consumption, thus affecting the subsequent feature selection and classification. To solve the problem and improve classification performance, a novel “three-step” method NLF-DMIC is proposed, which improves classification accuracy and computational efficiency to a large extent.

NLF-DMIC selects features on missing data by PMIC at first, then fills the missing values on the selected feature subset by NLF model, and finds out the best feature set by LMIC at last. A large number of experiments on a synthetic dataset and 8 real datasets proved that, in comparison to “two-step”, the “three-step” method NLF-DMIC can find a feature subset with higher classification accuracy and reduce computational time.

The main advantages of NLF-DMIC are (1) the feature selection is divided into two steps, which takes into account that irrelevant features are not conducive to imputation but redundant features contribute to imputation. (2) the first “rough” feature selection eliminates most of the irrelevant features, reducing imputation errors and computational cost; (3) the proposed LMIC performs well in elimination of feature redundancy. However, a limitation of the “three-step” feature selection method is “partial sample strategy” cannot ensure the validity of feature selection at the high missing rate and a small number of samples. This is still a tough problem that deserves further study.

CRedit authorship contribution statement

Jianhua Hu: Methodology, Validation, Conceptualization, Formal analysis, Writing – review & editing. **Kejin Pan:** Methodology, Writing – original draft, Software, Visualization. **Yan Song:** Conceptualization, Writing – review & editing, Supervision. **Guoliang Wei:** Supervision, Validation, Funding acquisition. **Chungen Shen:** Data curation, Validation.

Data availability

Data will be made available on request.

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